

$$① f(x, y) = x^2 \operatorname{sen} y$$

$$\frac{\partial f}{\partial x} = 2x \operatorname{sen} y$$

$$\frac{\partial f}{\partial y} = x^2 \cos y$$

$$\frac{\partial f}{\partial x \partial y} = 2x \cos y$$

$\longleftrightarrow$

$$\frac{\partial f}{\partial y \partial x} = \cos y \cdot 2x$$

$$\frac{\partial^2 f}{\partial x^2} = 2 \operatorname{sen} y$$

$$\frac{\partial^2 f}{\partial y^2} = -x^2 \operatorname{sen} y$$

$$② f(x, y) = x^2 y \operatorname{sen} y$$

$$\frac{\partial f}{\partial x} = 2x y \operatorname{sen} y$$

$$\frac{\partial f}{\partial y} = x^2 (\operatorname{sen} y + y \cos y)$$

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$$\frac{\partial^2 f}{\partial x^2} = 2 y \operatorname{sen} y$$

$$\frac{\partial^2 f}{\partial y^2} = x^2 (\cos y + \cos y + y (-\operatorname{sen} y))$$

$$\frac{\partial^2 f}{\partial y^2} = x^2 (2 \cos y - y \operatorname{sen} y)$$

TABLA

$$y = u \cdot v$$

$$\rightarrow y' = u'v + uv'$$

$$y = \operatorname{sen} x$$

$$y' = \cos x$$

$$y = \cos x$$

$$y' = -\operatorname{sen} x$$